Project Proposal

Hybrid Retrieval-Augmented Generation with Parameter-Efficient Fine-Tuning for Customer Support Intent Classification and Response Generation

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Task 1.1 – Define Project Scope

1.1.1 Identify the Target Use Case

Target Use Case

This project **proposes to develop** a customer support assistant for **Refund Support**, where customers submit refund, cancellation, or return requests. The proposed system **will identify** the customer's intent, **retrieve** the relevant Standard Operating Procedure (SOP), and **generate** an accurate, policy-compliant response.

Expected Input

JSON
```json
{
  "query": "I received a damaged product and want a refund."
}
```

Expected Output

JSON
```json
{
```

```
  "intent": "refund_request",
  "category": "REFUND",
  "response": "According to the company refund policy, damaged products are eligible for a full refund within 30 days of delivery."
}
```

Project Scope

In Scope

- Processing refund-related customer queries.
- Intent classification using Parameter-Efficient Fine-Tuning (PEFT).
- SOP retrieval using Hybrid Retrieval-Augmented Generation (Hybrid RAG).
- Policy-grounded response generation.
- Structured JSON-based outputs.

Out of Scope

- Live order tracking.
- Payment processing.
- Customer authentication.
- Human agent escalation workflows.
- Multi-turn conversational memory.

1.1.2 Define Success Criteria

The proposed solution **will be considered successful** if it achieves the following evaluation targets.

Metric	Target
Format Adherence	100% valid JSON responses
ROUGE-1	≥ 0.70
ROUGE-L	≥ 0.70
BLEU	≥ 0.50
Hallucination Reduction	Lower than the Naive RAG baseline
Intent Classification	High accuracy on the held-out test set
Policy Compliance	Responses grounded in retrieved SOPs

Evaluation Objectives

- Generate 100% schema-compliant JSON outputs.
- Improve response quality compared with the baseline LLM and Naive RAG.
- Reduce hallucinations through Hybrid RAG and fine-tuned intent classification.
- Ensure responses are grounded in retrieved company policies.

Task 1.4 – Design Methodology

1.4.1 Select Baseline Model

The **proposed baseline model** for this project is **Qwen2.5-1.5B-Instruct**.

This model **is proposed** because it provides a good balance between performance, memory efficiency, and instruction-following capability. Its relatively small parameter size **is expected to enable** efficient fine-tuning using QLoRA on a single NVIDIA T4 GPU (16 GB VRAM) while maintaining competitive performance for customer support tasks. The model **is expected to support** structured JSON generation and intent classification, making it suitable for the Bitext customer support dataset.

Selection Justification

- Optimized for instruction-following tasks.
- Supports efficient QLoRA fine-tuning on limited GPU resources.
- Suitable for structured JSON output generation.
- Computationally efficient compared with larger LLMs.

1.4.2 Design Retrieval Strategy

The proposed solution **will use** a Hybrid Retrieval-Augmented Generation (Hybrid RAG) pipeline to generate company-specific responses grounded in Standard Operating Procedures (SOPs).

Embedding Model

- Sentence-Transformers: **all-MiniLM-L6-v2**

Vector Database

- Chroma DB

Retrieval Workflow

1. Customer queries **will be classified** into intent and support category.
2. The predicted intent **will be mapped** to a retrieval query.
3. The query **will be converted** into vector embeddings.
4. Chroma DB **will perform** cosine similarity search to retrieve the most relevant SOP chunk.
5. The retrieved context **will be appended** to the final prompt before response generation.

Similarity Search

- Cosine similarity
- Top-1 document retrieval ($k = 1$)

Retrieval Evaluation

The effectiveness of retrieval **will be assessed** by verifying whether the retrieved SOP chunk is relevant to the predicted customer intent. The overall response quality **will be evaluated** using:

- ROUGE-1
- ROUGE-L
- BLEU
- Policy grounding against retrieved SOP content

1.4.3 Design Fine-Tuning Strategy

Parameter-efficient fine-tuning **will be performed** using QLoRA (Quantized Low-Rank Adaptation) to reduce GPU memory usage while adapting the model for customer support intent classification.

Fine-Tuning Strategy

- Method: QLoRA
- Quantization: 4-bit NF4
- PEFT Framework: LoRA

Target Modules

LoRA adapters **will be applied** to the following attention projection layers:

- `q_proj`
- `k_proj`
- `v_proj`
- `o_proj`

Training Format

Training data **will be converted** into ChatML format consisting of:

- System message
- User instruction
- Assistant response containing structured JSON

Checkpointing

- The training pipeline **will save** the best model checkpoints.
- Tokenizer and adapter weights **will be stored** for inference.

Experiment Tracking

Training progress and evaluation metrics **will be tracked** using MLflow.

1.4.4 Define Evaluation Framework

The proposed solution **will be evaluated** using multiple complementary metrics.

Metric	Purpose
Format Adherence	Measures whether responses follow the required JSON schema.
Intent Accuracy	Evaluates the correctness of predicted intent and category.
Output Consistency	Measures whether identical queries produce consistent outputs.
Hallucination Frequency	Measures the percentage of responses containing information not supported by the retrieved SOP.
ROUGE-1 / ROUGE-L	Measures lexical similarity between generated and reference responses.
BLEU	Measures n-gram similarity between generated and reference responses.

The evaluation framework **will compare** the following approaches:

- Baseline LLM
- Naive RAG
- Hybrid RAG (Fine-Tuned Intent Router + Retrieval)